

BEYOND RPA • ADVANCE AGENTIC AUTOMATION

# Advance Agentic RPA & UiPath Architecture

A 30-session masterclass covering Document Intelligence, AI Center ML pipelines, Studio Autopilot, Self-Healing Selectors, Model Context Protocol (MCP) Servers, Python Coded Agents, and UiPath Maestro Multi-Agent Swarms.

## THE PARADIGM SHIFT: BEYOND TRADITIONAL RPA

Transition from brittle screen scraping and fixed flowchart bots to autonomous reasoning agents equipped with dynamic tool registries, validation guardrails, and enterprise BPMN orchestration.

SESSIONS

**30**

Live interactive builds

MODULES

**12**

Enterprise architecture

PHASES

**4**

Structured progression

FORMAT

**100%**

Hands-on code repos

[Explore Course Webpage →](#)[Talk to RPA Vault Mentors →](#)

CURRICULUM ARCHITECTURE

# Beyond RPA: *Architecture for the Modern AI Stack*

Stop training for the automation platform of two years ago. Master the full stack of intelligent document understanding, reasoning agents, and MCP protocols.

SESSIONS  
**30 Live**  
Code builds

30

MODULES  
**12 Deep**  
Full stack

12

PHASES  
**4 Stages**  
Zero to agentic

04

BUILDS  
**100% Code**  
Git repos

100%

THE PAST  
**Where Most Courses Stop**

01

Legacy RPA methodologies causing production maintenance nightmares:

- **Brittle Selectors:** Flows break on DOM updates.
- **Hardcoded Regex:** Inflexible rules on variable layouts.
- **Manual Try/Catch:** Linear error traps without LLMs.
- **Siloed Bots:** Incapable of multi-agent swarm flows.

THE FUTURE  
**What You Build in This Track**

02

Production-grade Agentic systems built for modern enterprise scale:

- **Autonomous LLM Agents:** Grounding & system prompts.
- **Model Context Protocol:** Tool APIs for LLMs.
- **Self-Healing Selectors:** ScreenPlay vision runtime.
- **Maestro BPMN Swarms:** Multi-agent coordination.

AUDIENCE  
**Who This Is Designed For**

03

Tailored specifically for engineers seeking senior career acceleration:

- **UiPath Developers:** Moving past screen scraping.
- **RPA Architects:** Designing enterprise AI stacks.
- **Senior Engineers:** Targeting ₹12-20L+ packages.

PREREQUISITES  
**What You Need & Prerequisites**

04

Zero AI/ML background needed — all concepts taught from first principles:

- **Studio Basics:** Familiarity with variables & flows.
- **Zero ML Required:** AI Center taught from scratch.
- **Python FastMCP:** Practical scripting taught step-by-step.

THE AGENTIC ADVANTAGE

**Build systems that reason, heal, and orchestrate across the modern AI stack.**

Learn Document Intelligence · Deploy AI Center ML models · Build Python MCP Servers · Orchestrate Multi-Agent Swarms with Maestro.

PHASE 01 • CORE FOUNDATIONS

# 4-Stage Document Understanding *Framework*

Master document digitization, intelligent taxonomy schemas, and production invoice extraction pipelines.

**Day 01 DU Fundamentals & Enterprise Document Types**

Structured, semi-structured, and unstructured document ingestion. OCR digitizer engines comparison (OmniPage, Google Cloud OCR, Microsoft OCR) and enterprise DU ROI.

**Day 02 The 4-Stage DU Framework & Taxonomy Manager**

Deep dive into Taxonomy Manager schema definitions, document type hierarchies, Digitization OCR engines, Keyword/Intelligent Classifiers, and Extractor orchestration.

**Day 03 End-to-End Production Invoice Processing Capstone** **CAPSTONE BUILD**

Full practical build implementing automated invoice processing, line-item table extractions, regex rule validation, exception routing, and enterprise debugging.

**Day 04 Studio Document Understanding Framework Template**

Understanding and customizing the enterprise Document Understanding framework template inside UiPath Studio, config asset integration, and state machine transitions.

**Day 05 Introduction to AI Center & ML Training**

Create and deploy ML packages, building dataset pipelines and training documents in Document Manager. Creating pipelines, ML Skill deployment, and retraining models.

**01 Taxonomy**

Schema mapping & field grouping JSON.

**02 Digitization**

Multi-engine OCR text & DOM parsing.

**03 Classification**

Keyword & Intelligent vector routing.

**04 Extraction**

Pre-trained ML & line-item parsing.

**PRACTICE LAB & CHECKPOINT:** Build a complete Document Understanding pipeline with taxonomy schema definition, OCR digitizer calibration, and fallback regex extraction rules.

Have questions about Document Understanding? Connect with our mentors

[Schedule Mentor Call →](#)

PHASE 01 • ENTERPRISE MODULES

Action Center HITL, *Cloud APIs & IXP*

Deploy modern pre-trained models, orchestrate Action Center HITL validation, connect Cloud APIs, and build IXP & Studio Web API workflows.

DAYS 06-07

06-07

Action Center HITL & Modern DU Models

Deploying human-in-the-loop validation tasks and leveraging modern pre-trained models for zero-shot extraction:

Day 06

Action Center & Human-in-the-Loop

Human-in-the-loop validation concepts, creating validation tasks, Action Center user routing workflows, and Orchestrator catalogs.

Day 07

Modern DU Architecture & Pre-Trained Models

Comparing traditional DU vs Modern DU, utilizing out-of-the-box pre-trained models (Invoices, Receipts, POs), and building end-to-end custom scenarios.

Action Center HITL Tasks Modern DU Pre-Trained Models

DAYS 08-10

08-10

Cloud APIs, IXP & Studio Web API Workflows

Integrating serverless Cloud APIs, next-gen Intelligent Xtraction, and API-driven automation flows:

Day 08

DU Cloud APIs

Consume Document Understanding capability using cloud REST APIs for high-throughput, serverless document extraction pipelines.

Day 09

Introduction to IXP (Intelligent Xtraction)

What IXP is, why it is required for complex layouts, and IXP vs traditional extraction comparison. Build an end-to-end production use case.

Day 10

Introduction to API Workflows

Understanding API workflows in Studio Web. Build a complete automated use case handling enterprise IT support tickets.

Cloud REST APIs IXP Framework Studio Web API Workflows

Action Center

Orchestrator catalogs & validation tasks.

IXP Next-Gen

Multi-page complex layout parsing.

API Workflows

Cloud triggers & Studio Web tickets.

**PRACTICE LAB & EXIT CHECK:** Deploy an Action Center HITL validation task with Modern DU pre-trained models, integrate Cloud APIs, and trigger downstream Studio Web API workflows on completion.

Want to see a live demo of AI Center & IXP pipelines?

Request Demo Access →

PHASE 02 • ADVANCED AI INSIDE UIPATH

# GenAI, Autopilot, *Self-Healing & ScreenPlay*

Master Generative AI activities, Autopilot for Everyone, dynamic Self-Healing Selectors, and ScreenPlay vision automation.

DAY 11

11

## GenAI Activities & Autopilot for Everyone

Harnessing Generative AI natively inside UiPath Studio and business operations:

Day 11

### GenAI Activities & Autopilot for Everyone

Introduction to GenAI activities. Deploy Autopilot for Everyone for enterprise business use cases. Use Context using Autopilot for Everyone.

Gen AI Activities

Autopilot for Everyone

Context Grounding

Studio AI

DAYS 12-13

12-13

## Self-Healing Selectors & ScreenPlay Vision

Building resilient automations that automatically recover from broken UI and navigate remote environments:

Day 12

### Introduction to Self-Healing Selectors

Dynamic selector recovery, self-healing runtime execution, and zero maintenance when web and desktop interfaces change.

Day 13

### ScreenPlay & Computer Vision

Next-generation vision-based automation patterns for automating challenging remote desktop and virtual environments.

Self-Healing Selectors

ScreenPlay Vision

Runtime Recovery

Remote Desktop AI

### Dynamic UI Healing

Fuzzy matching & visual anchor backup recovery.

### Autopilot Studio

Natural language workflow & expression synthesis.

### ScreenPlay Vision

Computer vision automation for Citrix & remote desktops.

**PRACTICE LAB & EXIT CHECK:** Deploy a self-healing bot utilizing ScreenPlay vision that automatically recovers from simulated UI element updates and passes semantic parameters to Gen AI activities.

Upgrade your UiPath skillset with Agentic AI and ScreenPlay Vision

Apply on Course Webpage →

## PHASE 03 • AUTONOMOUS AGENTS

# Agent Builder, Guardrails & *AgentScore Matrix*

Build reasoning agents in UiPath Agent Builder, define system prompt guardrails, and evaluate agent performance with Agent Score.

DAYS 14-17

14-17

## Agent Builder & Prompt Guardrails Architecture

Transitioning from deterministic automation to autonomous decision loops with structured system instructions and memory context:

Day 14

### Introduction to LLM and Agentic AI

What an LLM is, LLM architecture, LLM vs Agentic AI, agent vs traditional automation architecture, and autonomous decision loops.

Day 15

### Agent Builder Core Concepts

Agent Builder core capabilities: System Prompt, User Prompt, Tool Calling, context grounding, and memory integration.

Day 16

### Build Your First Agent

Live production build configuring autonomous agent parameters, input/output schemas, and testing.

Day 17

### Prompt Engineering & Guardrails

Prompt engineering techniques (Zero-shot, Few-shot, Prompt Chaining), schema validation, and practical guardrails implementation.

Agent Builder

System Instructions

Few-Shot Prompting

Guardrails

DAYS 18-21

18-21

## RAG, HITL Escalations & AgentScore Benchmarking

Governing agent behavior in production: context grounding, Action Center human-in-the-loop, and measuring reasoning accuracy:

Day 18

### Context Grounding and RAG

Introduction to Context Grounding. What RAG is and how it works with LLMs. End-to-end production use case on RAG.

Day 19

### Agent Escalations (HITL) & Action Center

Create Action APP for human validation, escalation thresholds, approval workflows, and Action Center integration.

Day 20

### Agent Evaluations & Agent Score

Evaluation benchmarks, metrics, ground truth datasets, testing agents, and calculating Agent Score.

Day 21

### Build an Agent with Tool Selection

End-to-end production use case configuring autonomous agents with intelligent dynamic tool selection.

RAG &amp; Grounding

Agent Score

HITL Action Center

Dynamic Tool Selection

### System Guardrails

Strict schema validation & hallucination guards.

### AgentScore Matrix

Ground-truth 50-item evaluation benchmarks.

### HITL Routing

Action Center escalation on low confidence.

**PRACTICE LAB & EXIT CHECK:** Create an autonomous Customer Escalation Agent with custom prompt guardrails, measure its Agent Score over a 50-item evaluation dataset, and trigger Action Center tasks on uncertainty.

Learn how to build production-grade Agent Builder systems

[Talk to an RPA Architect →](#)

PHASE 04 • FLAGSHIP LAYER

# MCP Servers, Python Coded Agents & *Maestro Swarms*

Build Model Context Protocol (MCP) servers in Python, code custom agents, and orchestrate multi-agent swarms with UiPath Maestro.

MODULE 01 • DAYS 23-27

23-27

## The Model Context Protocol (MCP) Server Suite

Mastering Anthropic's open standard for connecting LLMs to enterprise tools across UiPath, Python, and local CLIs.

Day 23 Introduction to Model Context Protocol (MCP)

What Model Context Protocol is, why it matters for autonomous agents, and client-server tool registration architecture.

Day 24 UiPath MCP Server Configuration & Tool Registration

Practical demo configuring and registering enterprise automation tools inside the native UiPath MCP Server.

Day 25 Coded MCP Server Development with Python + UiPath

Practical demo building a custom Coded MCP Server using Python + UiPath to expose custom business logic to LLMs.

Day 26 Command MCP Server for Local CLI & Process Tools

Local tool execution, process invocation, and command-line Model Context Protocol integration.

Day 27 Remote MCP Server Enterprise Deployment & Security

Distributed enterprise MCP server deployment, remote endpoint hosting, and network security protocols.

MODULE 02 • DAYS 22 & 30

22 & 30

## Python Coded Agents & AI Coding

Writing pure Python automation scripts, deploying to Orchestrator, and synthesizing workflows using Claude 3.5 Sonnet & Codex.

Day 22 Python Coded Agents

Building a Coded Agent in Python and deploying to Orchestrator.

Day 30 AI Coding Skills (Claude & Codex)

Synthesizing and optimizing RPA workflows automatically.

MODULE 03 • DAYS 28-29

28-29

## UiPath Maestro & CrewAI Swarms

BPMN 2.0 process modeling, state management, event orchestration, and coordinating autonomous CrewAI swarms.

Day 28 Maestro BPMN 2.0 Flows

BPMN standard, tasks, events, and decision gateways.

Day 29 Multi-Agent Swarm Demo

Executing swarms with external CrewAI frameworks.

**FLAGSHIP CAPSTONE ARCHITECTURE PROJECT:** Architect an end-to-end autonomous enterprise system connecting a Python FastMCP Server, an Agent Builder reasoning agent, and a UiPath Maestro BPMN gateway for end-to-end order fulfillment.

The flagship layer separating you from the market

[View Full Capstone Details →](#)

## WHAT'S INCLUDED • LIVE DELIVERY

**Everything you need to master *next-generation automation***

Live interactive coding, production codebases, 1-on-1 mentor code reviews, and industry-recognized credentials.

## LEAD INSTRUCTOR

01

**Akash Javalekar****UiPath RPA Tech Lead | AI & Agentic Automation Expert**

Consultant @ Capgemini · UiPath Community Champion · 5+ Yrs Exp

- 🏆 UiPath Community Automation Champion
- ★ Featured on the UiPath Champion Page
- 🏆 6× UiPath Community Forum Award Winner
- 🧠 AI & Agentic + Document Understanding Specialist

## PILLAR 02

02

**Production-Grade Architecture**

Build real Document Understanding pipelines, AI Center ML models, Python MCP servers, Coded Agents, and Maestro BPMN flows.

- Enterprise design patterns & standards
- Production error handling & guardrails

## PILLAR 03

03

**Lifetime Recording & Code Access**

Full HD recordings of every class with complete project source code repositories for continuous reference and production reuse.

- Full HD video recordings & timestamps
- Clean GitHub repositories for all modules

## PILLAR 04

04

**Direct Mentor Code Review**

1-on-1 architecture guidance and pull-request reviews on your agent workflows from practicing RPA and AI architects.

- Personalized 1-on-1 feedback sessions
- Direct Git pull-request code reviews

## PILLAR 05

05

**Private Practitioner Community**

Exclusive network of working RPA architects and Agentic AI developers sharing live job openings, project tips, and interview insights.

- Verified job leads & interview prep
- Architect networking & knowledge base

## PILLAR 06

06

**Industry-Recognized Credential**

Official Certificate of Completion recognized across top enterprise employers, consulting firms, and hiring partners.

- Verifiable digital certificate ID
- Showcase on LinkedIn & resume

## READY TO CLOSE THE GAP?

**Advance your career to the forefront of Agentic Automation.**

30 sessions. Four phases. One destination: agentic. Connect with RPA Vault mentors to enroll in the next live batch.

Limited batch sizes for personalized mentor code reviews

[Schedule Admissions Call →](#)

YOUR NEXT PRACTICAL STEP

# Learn the tools. Build the evidence.

Explore the complete course, review the 30-session curriculum, and connect with RPA Vault mentors to plan your advanced automation journey.

**Explore the complete course**

See the full curriculum, learning model, project path, and course details online.

[rpavault.com/course/advance-agentic-rpa-uipath/](https://rpavault.com/course/advance-agentic-rpa-uipath/) →

**Talk to RPA Vault Mentors**

Discuss your background, course fit, batch options, and enrollment steps.

[rpavault.com/contact/](https://rpavault.com/contact/) →

[Explore Course Webpage →](https://rpavault.com/course/advance-agentic-rpa-uipath/)[Talk to RPA Vault Mentors →](https://rpavault.com/contact/)